MILITARY SPECIFICATION

ANTENNA COUPLER GROUP AN/SRA-34A(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for Antenna Coupler Group AN/SRA-34A(V), an antenna coupling, radio frequency (RF) switching and adapting equipment which, in functioning as the connecting link between radio equipments such as the AN/SRC-23A and standard Navy radios, and shipboard antenna systems, will provide an overall compatible shipboard high frequency Communications Systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, unless otherwise specified herein, or request proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore; General Specification for
MIL-I-16910	- Interference Measurement, Electromagnetic, Methods and Limits,
MIL-W-21965	- Water Cooling of Shipboard Electronic Equipments; Specification for.
MIL-C-24056	- Cabinets, Electrical Equipment CY-4516/S (Single Bay) (Naval Shipboard use)

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-188	- Military Communication System Technical Standards

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MIL-STD-480	- Configuration Control Engineering Changes, Deviations and Waivers.
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-1399	- Interface Standard for Shipboard Systems

PUBLICATIONS

MILITARY

NAVSHIPS 93820	- Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The AN/SRA-34A(V) shall conform to the requirements of MIL-E-16400 for a shipboard production equipment, except as otherwise specified herein.

3.1.1 Environment. The AN/SRA-34A(V) shall conform to the Range 4 temperature requirements of MIL-E-16400.

3.1.2 Uniformity. The electronic assemblies of AN/SRA-34A(V) shall be electrically and physically interchangable with those of the Communication Central AN/SRC-16A to the maximum extent possible. Insofar as practical, the design of equipment shall employ a minimum of types and kinds of parts. Assemblies performing similar functions shall be indentical in design and construction.

3.1.3 Reliability. The equipment shall conform to the reliability test requirements of 4.5

3.1.4 Shock. The AN/SRA-34A(V) shall be capable of meeting the grade A, class I, type A shock requirements of MIL-S-901.

3.1.4.1 Locking devices. The use of locking devices on controls shall be as necessary.

3.1.5 Cooling. Water cooling for the equipment shall be as specified herein; water cooling in accordance with method 1 of MIL-W-21965 shall be provided, except that the maximum allowable heat dissipation to the local environment shall be 1000 watts or 25 percent of the total heat generated in each equipment cabinet, whichever is less.

3.1.6 Enclosure. The AN/SRA-34A(V) shall be splashproof enclosed in accordance with MIL-STD-108.

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3.1.7 Primary power. The AN/SRA-34A(V) shall be capable of operation within the requirements of this specification when connected to a 115 volt, 400 Hertz (Hz), 3 phase power source whose characteristics are as specified by MIL-STD-761 for type I systems.

3.1.7.1 Equipment utilization. The AN/SRA-34A(V) shall conform to the requirements specified by MIL-STD-761 type I systems, for utilization equipment.

3.1.8 Illumination. The illumination requirements of all dials, control and lights shall be governed by conditions of equipment installation in an illuminated compartment onboard ship.

3.1.9 Cabling. Cabling, loops, lines, and trunks shall be in accordance with MIL-STD-188.

3.1.10 Acoustical noise. Acoustical noise shall be held to an absolute minimum and shall not hinder normal operator function within the immediate area.

3.1.11 Blown fuse indicators. Wherever fuses are used for protection against damaging overloads, blown fuse indicators shall be provided.

3.1.12 Connectors. Required mating cable connectors shall be furnished as part of the AN/SRA-34A(V). Stuffing tubes, if used, shall also be furnished.

3.1.13 Integral mounting. When mounting in a CY-4516/S cabinet is specified for the AN/SRA-34(V), all requirements for the equipment specified herein shall be met while mounted in and as an integral part of the CY-4516/S cabinet specified in MIL-C-24056.

3.1.14 Electromagnetic interference emission and susceptibility. The AN/SRA-34A(V) shall withstand the tests of MIL-I-16910.

3.2 System requirements.

3.2.1 System operation. The AN/SRA-34A(V) in conjunction with standard Navy RF equipments, such as AN/SRT-15, AN/WRT-2, AN/URT-23(V) transmitters, AN/URC-32, Radio Set, and AN/SRC-23A or equivalent radio equipment shall be capable of providing up to eight RF channels in the 2 to 6 MegaHertz (MHz) range, eight RF channels in the 4 to 12 MHz range, and eight RF channels in the 10 to 30 MHz range. The use of any combination of the above radio equipments at full power output shall be possible.

3.2.1.1 Combined operation. Independent and simultaneous operation of any combination of simplex and full deplex RF circuits, within the restrictions of up to eight total RF channels per antenna separated on a frequency basis by a minimum of 5 percent transmitter-to-transmitter and 10 percent receiver-to transmitter shall be possible on one 2 to 6 shipboard broadband antennas.

3.2.2 The AN/SRA-34A(V) shall be designed so that there shall be substantially no disturbance to other transmitters during on-off and tuning periods. In addition, it shall not be necessary to disable any transmitter or receiver in order to connect, activate, tune, or adjust any other transmitter.

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3.2.3 Operational certification. The capability shall be provided to permit rapid certification that the system is operationally usable at any desired time by providing special test signals, unit samples and analyzing circuits capable of indicating the systems operational status when operating in the communication circuit or when in standby condition.

3.2.4 Tune-up. The antenna coupling networks shall be capable of tune-up without actual operation on the radiating antenna with an emission of less than 50 microwatts on a nominal 50-ohm antenna.

3.2.5 Automatic tuning. The use of automatic tuning, alignment and control of equipments shall be used to attain reduction in complexity, improved performance, reliability, and reasonable reductions in tuning times. Automatic tuning shall not be used for the sole purpose of subordinating operator's judgment. Manual overrides shall be provided wherever needed to allow for system operation despite a breakdown in any automatic control circuit.

3.2.6 Connection. Connection between RF equipments and the antenna couplers shall be made via the automatic RF Switching matrix, or a manual shipboard RF transfer panel not supplied as part of the AN/SRA-34A(V).

3.2.7 Flexibility. The AN/SRA-34A(V) shall contain sufficient flexibility to allow for variable configurations of multiple channel installations on a building block concept.

3.2.8 Interlock. Provisions shall be made to interlock the RF channels to insure RF power removal during the period when a matrix crosspoint is being closed or opened.

3.2.9 Adapters. The design of the transmitter adapters and coupler adapters shall be such as to allow full compatible operation of any combination of transmitters and couplers specified in 3.3.6 and 3.3.7.

3.2.10 Adapter circuitry. The adapters shall provide the necessary circuitry required to diplex on the RF coaxial and process all control and monitoring information provided in the equipments to allow for flexible patching of any combination of the equipments via the RF matrix (SA-1070/SRA-34) without restriction. The operation of any combination of couplers and transmitters shall be possible in all modes (including tuning) of operation inherently designed in each equipment and the tuning process shall be such as to provide protection against inadvertent equipment damage.

3.3 Equipment characteristics.

3.3.1 Group composition. The AN/SRA-34A(V) shall be composed of equipment in the form of the following basic building blocks in combinations and quantity as specified in the contract or order:

(a) Antenna coupler group (2 to 6 MHz) OA-4794A/SRA-34(V) consisting of:

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(1) Cabinet, electrical equipment; CY-4032A/SRA-34(V).
(2) Four CU-1169/SRC-16 (2 to 6 MHz) antenna couplers.

(b) Antenna coupler group (4 to 12 MHz) OA-( )/SRA-34(V) consisting of:

(1) Cabinet, electrical equipment; CY-( )/SRA-34(V).
(2) Four CU-( )/SRA-34(V) (4 to 12 MHz) antenna couplers.

(c) Antenna coupler group (10 to 30 MHz) OA-( )/SRA-34(V) consisting of:

(1) Cabinet, electrical equipment; CY-( )/SRA-34(V).
(2) Four CU-( )/SRA-34(V) (10 to 30 MHz) antenna couplers.

(d) Switching unit, R.F.: SA-1070/SRA-34(V).
(e) Electrical equipment cabinet (switching unit, cabinet): CY-4033/SRA-34(V).
(f) Control, remote switching: C-4787/SRA-34(V).
(g) Adapter, transmitter: MX-4847B/SRA-34(V).
(h) Adapter, coupler MX-4845A/SRA-34(V).

3.3.2 Antenna coupler group: OA-4794A/SRA-34(V). The coupler groups of 3.3.1(a), (b), and (c) shall be water cooled and housed in cabinet CY4032A/SRA-34(V) having dimensions (to the nearest inch) of 25 inches (63.50cm) in width, 28 inches (71.12cm) in depth and 72 inches (182.88cm) in height. The cabinet shall be so designed as to permit the operation of any combination of the three types of couplers up to a maximum of four couplers per cabinet.

3.3.2.1 Antenna coupler CU-1169/SRC-16. The coupler shall be capable of automatic tuning with manual override, and shall provide the proper impedance matching and selectivity between a transmitter (or receiver, or both) and the antenna in the 2 to 6 MHz frequency range.

3.3.2.1.1 Impedance. The coupler shall present a 50-ohms, unbalanced, 1.3 to 1 voltage standing wave ratio (VSWR) maximum, input impedance at the operating frequency. The load (antenna) impedance shall be 50 ohms unbalanced, with a 4 to 1 VSWR maximum.

3.3.2.1.2 Coupler power input. The coupler shall be capable of continuous operation at RF power inputs of up to 5 kilowatts (KW) (PEP) and 2500 watts (average) at a minimum efficiency of 60 percent.

3.3.2.1.3 Forward and reverse attenuation. The forward and reverse attenuation shall be 35 decibels (dB) at minus 5 and plus 5 percent of the operating frequency. The forward attenuation shall increase to a minimum of 45 dB and the reverse attenuation shall increase to a minimum of 50 dB at minus 15 and plus 15 percent of the operating frequency and shall not decrease outside the region plus or minus 15 percent of operating frequency throughout the tuning range of the coupler.

3.3.2.1.4 Phase stability. The phase stability of the couplers shall be such that the phase shift shall not exceed 1 degree in 22 milliseconds (ms) during steady state operating conditions.

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3.3.2.1.5 Tune-up cycle. During the antenna coupler tune-up cycle, the transmitter shall provide the unit a single tone of 100 watts minimum, 150 watts maximum forward power. The unit shall maintain constant surveillance of the coupler input impedance and not disrupt the transfer of forward power to the antenna for average forward powers above 100 watts by automatically maintaining the tuning accuracy necessary to meet the input impedance requirements of 3.3.3.1.1. The tuning time required shall be less than 60 seconds.

3.3.2.3 Cabinet CY-4032A/SRA-34(V). The cabinet shall contain the coder/decoder facilities required to diplex the required tune control and monitoring information on the RF cable for four couplers.

3.3.2.3.1 Time meter. One elapsed time meter shall be included.

3.3.3 Switching unit, RF SA-1070/SRA-34(V). The switching unit shall be a lattice type switching array (matrix) to allow the connection of up to five RF equipments to any of up to 25 antenna coupling networks. The unit shall be so designed to permit variable configuration of one to four units for maximum of 20 RF inputs by 25 coupler outputs.

3.3.3.1 Matrix. The matrix shall be capable of being operated and monitored locally as well as from controls remotely located (up to 900 feet).

3.3.3.2 Power handling. The RF power handling capability of each RF crosspoint and channel shall be up to 5 KW (PEP and Average).

3.3.3.3 Matrix interlock. The matrix shall be interlocked so that it will not be possible to close more than one crosspoint in one row or in one column of the matrix at one time.

3.3.3.4 Crosstalk. The crosstalk between channels shall be at least 80 dB below signal levels.

3.3.3.5 Impedance. The input and output impedances shall be 50 ohms nominal with an input VSWR not to exceed 1.3 to 1 when terminated in 50 ohms.

3.3.4 Size. The CY-4033/SRA-34(V) Cabinet shall not exceed 12 cubic feet, and shall have all the required facilities to house up to four RF switching units.

3.3.4.1 Mounting. The cabinet shall be configured for deck or shelf mounting.

3.3.4.2 Dummy load adapter. One dummy load adapter shall be included with each CY-4033/SRA-34(V) cabinet. In systems where CY-4033/SRA-34(V) cabinets are not procured, a dummy load adapter shall be provided.

3.3.5 Remote switching control C-4787/SRA-34(V). The control shall permit the remote operation and monitoring of the RF matrix.

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3.3.5.1 Flexibility. The unit shall be designed to permit variable configurations of one to four units paralleling that of the RF matrix.

3.3.5.2 Size. The unit shall not exceed 2 cubic feet in volume, and shall be configured for completely compatible mounting in a CY-4516/S cabinet (see 3.1.13).

3.3.6 Transmitter adapter MX-4847B/SRA-34(V). The adapter shall permit a standard Navy RF equipment, such as the AN/SRT-15, AN/WRT-2, AN/URT-23 and AN/URC-32, capable of a minimum of 125 watts RF power output, to be operated with a coupler in the antenna coupler group.

3.3.6.1 Adapter coder/decoder. The adapter shall contain a coder/decoder to process all necessary control and monitoring information diplexed on the RF coaxial line.

3.3.6.2 Size. The unit shall not exceed 0.75 cubic feet in volume, and shall be configured for completely compatible mounting in the CY-4516/S cabinet (see 3.1.13).

3.3.6.3 The adapter shall be capable of performing its function when located up to 50 feet (15.24m) from its associated Navy RF equipment.

3.3.7 Coupler adapter MX-4845A/SRA-34(V). The adapter shall permit a Navy antenna coupler or tuner such as AN/SRA-13, AN/SRA-14, AN/SRA-15, AN/SRA-16, AN/URA-38, AN/SRA-22 and TN-342/WRT-2 tuner to be operated with the automatically tuned receiver-transmitter group OA-8012A/SRC-23(V) or the standard Navy RF equipment operating in conjunction with a transmitter adapter (see 3.3.6).

3.3.7.1 Adapter coder/decoder. The adapter shall contain a coder/decoder to process all necessary control and monitoring information diplexed on the RF coaxial line.

3.3.7.2 Size. The unit shall not exceed 0.75 cubic feet in volume and shall be configured for completely compatible mounting in a CY-4516/S cabinet (see 3.1.13).

3.3.7.3 The adapter shall be capable of performing its function when located up to 25 feet (7.62m) from its associated Navy coupling equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First article inspection (4.3)
(b) Quality conformance inspection (4.4)

(1) Production inspection (4.4.1)
(2) Environmental inspection (4.4.2)

(c) Reliability testing (4.5)
(d) Inspection of preparation for delivery (4.4.3)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine conformance with the requirements of this specification. First article inspection shall include the examination and testing specified in production inspection, and environmental inspection.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of 3.2 and 3.3.

4.4.2 Environmental inspection. Environmental inspection shall be as listed below and shall be performed on units which have been subjected to and passed production inspection.

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Table I. Environmental inspection.

Inspection	Reference paragraph
Low temperature	3.1.1
High temperature	3.1.1
Steady state voltage and frequency	3.1.7
Transient voltage	3.1.7
Transient frequency	3.1.7

4.4.2.1 Sampling for environmental inspection. One equipment shall be selected from each successive 50 equipments produced or fraction thereof. The first sample equipment shall be selected from the first months production.

4.4.3 Inspection of preparation for delivery. Inspections shall be conducted to insure conformance with the requirements of Section 5 of this specification.

4.5 Reliability test. The equipment shall be subjected to a 700 hour reliability test in accordance with test plan XXVIII of MIL-STD-781. The test shall be in accordance with test level C of MIL-STD-781 except that the "off" cycle shall be one hour per day. Input voltage cycling shall be required.

4.5.1 Test sample. The total operating test time shall be 700 hours to be accumulated on one unit or devided between two or more units. The test shall be completed prior to the delivery of production equipment.

4.5.2 Failure. Failures are malfunctions or parameter deviations which prevents the equipment from performing within the operational requirements specified herein.

4.5.3 Evaluation of test. No specific accept/reject criteria is established. The contractor shall evaluate each observed failure to determine the cause. Where equipment manufacturer design or part manufacturer design deficiencies are identified or when pattern failures (as defined by MIL-STD-781) exist, the contractor shall propose and implement corrective action to prevent or reduce the probability of recurrence. The test shall be considered successfully completed when all corrective actions have been approved by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, packing, and marking shall be in accordance with MIL-E-17555.

5.2 Sub-contracted material and parts. The preparation for delivery requirements or referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6. NOTES

6.1 Intended use. This equipment is intended for shipboard use.

6.2 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.2.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the environmental inspection.

Preparing activity:
Navy-EC
(Project 5985-N431)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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